REFERENCE

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1. Setup

- To print n decimals: `cout << fixed << setprecision(n);`

Bash script for compilation and execution

setup/compilation.sh

```bash
co() {
    g++ -std=c++20 -O2 -Wall -Wextra -Wshadow -Wconversion -o $1 $1.cpp
}

run() {
    co $1 && ./$1;
}
```

C++ template

setup/template.h

```cpp
#include <bits/stdc++.h>

using namespace std;

#define endl '\n'
#define dbg(x) cout << #x << ": " << x << endl
#define sz(x) int((x).size())
#define all(v) v.begin(), v.end()
#define rall(v) v.rbegin(), v.rend()
#define pb push_back
#define mp make_pair
#define fi first
#define se second

using ll = long long;
const ll INF = 1e18;
const ll MOD = 1e9 + 7;
```

2. Estructuras de Datos

2.1. Prefix Sums

2.1.1. 2D Prefix Sums

To process Q queries for the sum over a subrectangle of a 2D matrix with N rows and M columns

$$\texttt{prefix}[a][b] = \sum_{i=1}^{a} \sum_{j=1}^{b} \texttt{arr}[i][j]$$

This can be calculated as follows for row index $1 \le i \le n$ and column index $1 \le j \le m$:

$$\texttt{prefix}[i][j] = \texttt{prefix}[i-1][j] + \texttt{prefix}[i][j-1] - \texttt{prefix}[i-1][j-1] + \texttt{arr}[i][j]$$

The submatrix sum between rows a and A and columns b and B, can thus be expressed as follows:

$$\sum_{i=a}^{A} \sum_{j=b}^{B} \texttt{arr}[i][j] = \texttt{prefix}[A][B] - \texttt{prefix}[a-1][B] - \texttt{prefix}[A][b-1] + \texttt{prefix}[a-1][b-1]$$

2.2. Segment tree

Segment tree

TIME
- $O(n)$ for building
- $O(\log n)$ for queries and updates

estructuras/segment-tree.cpp

```cpp
int n, t[4*MAXN];

void build(int a[], int v, int tl, int tr) {
    if (tl == tr) {
        t[v] = a[tl];
    } else {
        int tm = (tl + tr) / 2;
        build(a, v*2, tl, tm);
        build(a, v*2+1, tm+1, tr);
        t[v] = t[v*2] + t[v*2+1];
    }
}
```

```cpp

int sum(int v, int tl, int tr, int l, int r) {
    if (l > r)
        return 0;
    if (l == tl && r == tr) {
        return t[v];
    }
    int tm = (tl + tr) / 2;
    return sum(v*2, tl, tm, l, min(r, tm))
        + sum(v*2+1, tm+1, tr, max(l, tm+1), r);
}

void update(int v, int tl, int tr, int pos, int new_val) {
    if (tl == tr) {
        t[v] = new_val;
    } else {
        int tm = (tl + tr) / 2;
        if (pos <= tm)
            update(v*2, tl, tm, pos, new_val);
        else
            update(v*2+1, tm+1, tr, pos, new_val);
        t[v] = t[v*2] + t[v*2+1];
    }
}
```

2.3. Policy-based data structures

The g++ compiler also supports some data structures that are not part of the C++ standard library. Such structures are called policy-based data structures. To use these structures, the following lines must be added to the code:

```cpp
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;
```

2.3.1. Ordered Set

We can define a data structure **indexed_set** that is like set but can be indexed like an array. The definition for int values is as follows:

```cpp
using ordered_set = tree<
    int,
    null_type,
    less<int>,
    rb_tree_tag,
    tree_order_statistics_node_update
>;
```

This data structure supports all the operations as a set in C++ in addition to the following:

- `find_by_order(k)`: returns an iterator to the k-th smallest element (0-based indexing)
- `order_of_key(x)`: returns the number of elements in the set that are strictly less than x

3. Ordenamiento y Búsqueda

3.1. Two Pointers

Some uses:

Subarray sum: Given an array of n positive integers and a target sum x, and we want to find a subarray whose sum is x or report that there is no such subarray.

2SUM Problem: Given an array of n numbers and a target sum x, find two array values such that their sum is x, or report that no such values exist.

Merging two sorted arrays: Given two arrays, sorted in ascending order combine the elements of these arrays into one big array.

```
while i < a.size() || j < b.size():
  if a[i] < b[j]:
    c[i + j] = a[i]
    i++
  else:
    c[i + j] = b[j]
    j++
```

Number of Smaller: We have two arrays a and b. We want to calculate for each element b_j how many such i exist that $a_i < b_j$.

```
i = 0
for j = 0..b.size()-1:
  while i < a.size() && a[i] < b[j]:
    i++
  res[j] = i
```

3.2. Binary Search on Monotonic Functions

Finding The Maximum x Such That $f(x)$ = true

TIME

$O(\log n)$

ordenamiento_busqueda/last_true.cpp

```cpp
#include <bits/stdc++.h>
using namespace std;

int last_true(int lo, int hi, function<bool(int)> f) {
    // if none of the values in the range work, return lo - 1
    lo--;
    while (lo < hi) {
        // find the middle of the current range (rounding up)
        int mid = lo + (hi - lo + 1) / 2;
        if (f(mid)) {
            // if mid works, then all numbers smaller than mid also work
            lo = mid;
        } else {
            // if mid does not work, greater values would not work either
            hi = mid - 1;
        }
    }
    return lo;
}

int main() {
    // all numbers satisfy the condition (outputs 10)
    cout << last_true(2, 10, [](int x) { return true; }) << endl;

    // outputs 5
    cout << last_true(2, 10, [](int x) { return x * x <= 30; }) << endl;

    // no numbers satisfy the condition (outputs 1)
    cout << last_true(2, 10, [](int x) { return false; }) << endl;
}
```

Finding The Minimum x Such That $f(x)$ = true

TIME

$O(\log n)$

ordenamiento_busqueda/first_true.cpp

```cpp
#include <bits/stdc++.h>
using namespace std;

int first_true(int lo, int hi, function<bool(int)> f) {
    hi++;
    while (lo < hi) {
        int mid = lo + (hi - lo) / 2;
        if (f(mid)) {
            hi = mid;
        } else {
            lo = mid + 1;
        }
    }
    return lo;
}

int main() {
    // outputs 2
    cout << first_true(2, 10, [](int x) { return true; }) << endl;
    // outputs 6
    cout << first_true(2, 10, [](int x) { return x * x >= 30; }) << endl;
    // outputs 11
    cout << first_true(2, 10, [](int x) { return false; }) << endl;
}
```

4. Teoría de gráficas

4.1. Graph Traversal

4.1.1. Floodfill

Flood fill is an algorithm that identifies and labels the connected component that a particular cell belongs to in a multidimensional array.

> ⚠ **Advertencia**
>
> Recursive implementations of flood fill sometimes lead to
> - Memory limit exceeded if you run the recursive implementation on a really big grid (i.e., running the above code on a 4000×4000 grid may exceed 256 MB)
>   - Non-recursive implementations generally use less memory than recursive ones.

Example (recursive implementation)

graficas/floodfill_recursive.cpp

```cpp
const int MAX_N = 1000;

int grid[MAX_N][MAX_N];  // the grid itself
int row_num;
int col_num;
bool visited[MAX_N][MAX_N];  // keeps track of which nodes have been visited
int curr_size = 0;           // reset to 0 each time we start a new component

void floodfill(int r, int c, int color) {
    if ((r < 0 || r >= row_num || c < 0 || c >= col_num)  // if out of bounds
        || grid[r][c] != color                             // wrong color
        || visited[r][c]                                   // already visited this square
    )
        return;

    visited[r][c] = true;  // mark current square as visited
    curr_size++;           // increment the size for each square we visit

    // recursively call flood fill for neighboring squares
    floodfill(r, c + 1, color);
    floodfill(r, c - 1, color);
    floodfill(r - 1, c, color);
    floodfill(r + 1, c, color);
}

int main() {
    /*
     * input code and other problem-specific stuff here
     */
    for (int i = 0; i < row_num; i++) {
        for (int j = 0; j < col_num; j++) {
            if (!visited[i][j]) {
                curr_size = 0;
                /*
                 * start a flood fill if the square hasn't already been visited,
                 * and then store or otherwise use the component size
                 * for whatever it's needed for
                 */
                floodfill(i, j, grid[i][j]);
            }
        }
    }
    return 0;
}
```

Example (iterative implementation)

graficas/floodfill_iterative.cpp

```cpp
#include <bits/stdc++.h>

using namespace std;

const int MAX_N = 2500;
const int R_CHANGE[]{0, 1, 0, -1};
const int C_CHANGE[]{1, 0, -1, 0};

int row_num;
int col_num;
string building[MAX_N];
bool visited[MAX_N][MAX_N];

void floodfill(int r, int c) {
```

```cpp
    // Note: you can also use a queue and pop from the front for a BFS-based
    // approach
    stack<pair<int, int>> frontier;
    frontier.push({r, c});
    while (!frontier.empty()) {
        r = frontier.top().first;
        c = frontier.top().second;
        frontier.pop();

        if (r < 0 || r >= row_num || c < 0 || c >= col_num || building[r][c] == '#' ||
            visited[r][c])
            continue;

        visited[r][c] = true;
        for (int i = 0; i < 4; i++) {
            frontier.push({r + R_CHANGE[i], c + C_CHANGE[i]});
        }
    }
}

int main() {
    cin >> row_num >> col_num;
    for (int i = 0; i < row_num; i++) { cin >> building[i]; }

    int room_num = 0;
    for (int i = 0; i < row_num; i++) {
        for (int j = 0; j < col_num; j++) {
            if (building[i][j] == '.' && !visited[i][j]) {
                floodfill(i, j);
                room_num++;
            }
        }
    }
    cout << room_num << endl;
}
```

4.2. Disjoin Set Union

Disjoint Set Union

TIME

- **find**: $O(\log n)$
- Con path compression y union by size/rank: $O(\alpha(n))$, donde α es la inversa de la función de Ackermann (amortizada).
- Con union by size/rank, peros sin path compression: $O(\log n)$ por consulta.

graficas/dsu.cpp

```cpp
#include "setup/template.h"

struct DSU {
    vector<int> parent, rank, size;
    int components;

    DSU(int n) : parent(n + 1), rank(n + 1, 0), size(n + 1, 1), components(n) {
        iota(all(parent), 0);
    }

    // Retorna el representante de la componente de "x"
    int find(int x) {
        return parent[x] = (parent[x] == x ? x : find(parent[x]));
    }

    // Retorna si "x" y "y" se encuentran en la misma componente
    bool connected(int x, int y) {
        return find(x) == find(y);
    }

    // Retorna el tamaño de la componente de "x"
    int get_size(int x) {
        return size[find(x)];
    }

    // Retorna la cantidad de componentes
    int count() {
        return components;
    }

```

```cpp
    // Une dos componentes y retorna el nuevo representante o
    retorna -1 si ya son parte de la misma componente
    int merge(int x, int y) {
        x = find(x);
        y = find(y);

        if (x == y) return -1;

        components--;

        if (rank[x] > rank[y]) swap(x, y);
        parent[x] = y;
        size[y] += size[x];

        if (rank[x] == rank[y]) rank[y]++;
        return y;
    }
};
```

4.3. Topological Sort

A topological sort of a directed acyclic graph is a linear ordering of its vertices such that for every directed edge $u \to v$ from vertex u to vertex v, u comes before v in the ordering.

DFS Version

graficas/topological_sort_dfs.cpp

```cpp
#include <bits/stdc++.h>

using namespace std;

vector<int> top_sort;
vector<vector<int>> graph;
vector<bool> visited;

void dfs(int node) {
    for (int next : graph[node]) {
        if (!visited[next]) {
            visited[next] = true;
            dfs(next);
        }
    }
    top_sort.push_back(node);
}

int main() {
    int n, m;  // The number of nodes and edges respectively
    cin >> n >> m;

    graph = vector<vector<int>>(n);
    for (int i = 0; i < m; i++) {
        int a, b;
        cin >> a >> b;
        graph[a - 1].push_back(b - 1);
    }

    visited = vector<bool>(n);
    for (int i = 0; i < n; i++) {
        if (!visited[i]) {
            visited[i] = true;
            dfs(i);
        }
    }
    reverse(top_sort.begin(), top_sort.end());

    vector<int> ind(n);
    for (int i = 0; i < n; i++) { ind[top_sort[i]] = i; }

    // Check if the topological sort is valid
    bool valid = true;
    for (int i = 0; i < n; i++) {
        for (int j : graph[i]) {
            if (ind[j] <= ind[i]) {
                valid = false;
                goto answer;
            }
        }
    }
answer:;

    if (valid) {
```

```cpp
        for (int i = 0; i < n - 1; i++) { cout << top_sort[i] + 1 << ' '; }
        cout << top_sort.back() + 1 << endl;
    } else {
        cout << "IMPOSSIBLE" << endl;
    }
}
```

BFS Version (Kahn's Algorithm)

graficas/topological_sort_bfs.cpp

```cpp
#include <bits/stdc++.h>

using namespace std;

int main() {
    int n, m;
    cin >> n >> m;

    vector<vector<int>> graph(n);
    for (int i = 0; i < m; i++) {
        int a, b;
        cin >> a >> b;
        graph[a - 1].push_back(b - 1);
    }

    vector<int> in_degree(n);
    for (const vector<int> &nodes : graph) {
        for (int node : nodes) { in_degree[node]++; }
    }

    queue<int> queue;
    for (int i = 0; i < n; i++) {
        if (in_degree[i] == 0) { queue.push(i); }
    }
    vector<int> top_sort;
    while (!queue.empty()) {
        int curr = queue.front();
        queue.pop();
        top_sort.push_back(curr);
        for (int next : graph[curr]) {
            if (--in_degree[next] == 0) { queue.push(next); }
        }
    }

    if (top_sort.size() == n) {
        for (int i = 0; i < n - 1; i++) { cout << top_sort[i] + 1 << ' '; }
        cout << top_sort.back() + 1 << endl;
    } else {
        cout << "IMPOSSIBLE" << endl;
    }
}
```

4.4. Lowest Common Ancestor (LCA)

Lowest Common Ancestor - Binary Lifting

graficas/lca.cpp

```cpp
#include<bits/stdc++.h>
using namespace std;

const int N = 3e5 + 9, LG = 18;

// adj[u] = lista de adyacencia del nodo u
vector<int> adj[N];
// up[i][j] = 2^j-ésimo ancestro de i
int up[N][LG + 1];
// depth[i] = profundidad del nodo i
int depth[N];
// subtree_size[i] = tamaño del subárbol con raíz en i
int subtree_size[N];

void dfs(int u, int p = 0) {
    up[u][0] = p;
    depth[u] = depth[p] + 1;
    subtree_size[u] = 1;

    for (int i = 1; i <= LG; i++) {
        up[u][i] = up[up[u][i - 1]][i - 1];
    }

```

```cpp
    for (auto v: adj[u]) {
        if (v != p) {
            dfs(v, u);
            subtree_size[u] += subtree_size[v];
        }
    }
}

// LCA de u y v
int lca(int u, int v) {
    if (depth[u] < depth[v]) swap(u, v);

    for (int k = LG; k >= 0; k--) {
        if (depth[up[u][k]] >= depth[v]) u = up[u][k];
    }

    if (u == v) return u;

    for (int k = LG; k >= 0; k--) {
        if (up[u][k] != up[v][k]) u = up[u][k], v = up[v][k];
    }

    return up[u][0];
}

// k-ésimo ancestro de u, k >= 0
int kth(int u, int k) {
    assert(k >= 0);
    for (int i = 0; i <= LG; i++){
        if (k & (1 << i)) u = up[u][i];
    }
    return u;
}

// Distancia entre u y v
int dist(int u, int v) {
    int l = lca(u, v);
    return depth[u] + depth[v] - (depth[l] << 1);
}

// k-ésimo nodo en el camino de u a v, el nodo 0 es u
int go(int u, int v, int k) {
    int l = lca(u, v);
    int d = depth[u] + depth[v] - (depth[l] << 1);
    assert(k <= d);
    if (depth[l] + k <= depth[u]) return kth(u, k);
    k -= depth[u] - depth[l];
    return kth(v, depth[v] - depth[l] - k);
}
```

5. Programación Dinámica

Kadane's algorithm

TIME
$O(n)$

Given an array of integers, find the maximum sum subarray.

dp/kadane.cpp

```cpp
long long kadane(vector<long long>& arr) {
   long long max_ending_here = arr[0];
   long long max_so_far = arr[0];
   for (size_t i = 1; i < arr.size(); ++i) {
       max_ending_here = max(arr[i], max_ending_here + arr[i]);
       max_so_far = max(max_so_far, max_ending_here);
   }
   return max_so_far;
}
```

Knapsack 0/1

TIME
$O(nW)$

Given weights and values of n items, put these items in a knapsack of capacity W to get the maximum total value in the knapsack.

dp/knapsack01.cpp

```cpp
int knapsack01(vector<int>& w, vector<int>& v, int W) {
   int n = w.size();
   vector<vector<int>> dp(n + 1, vector<int>(W + 1, 0));

```

```cpp

    for (int i = 1; i <= n; ++i) {
        for (int j = 0; j <= W; ++j) {
            dp[i][j] = dp[i-1][j]; // no tomar el objeto i
            if (w[i-1] <= j)
                dp[i][j] = max(dp[i][j], dp[i-1][j - w[i-1]] +
                v[i-1]);
        }
    }
    return dp[n][W];
}
```

Coin Change

dp/coin_change.cpp

```cpp
long long coin_change(vector<int>& coins, int X) {
    vector<long long> dp(X + 1, 0);
    dp[0] = 1; // una forma de formar 0

    for (int c : coins) {
        for (int x = c; x <= X; ++x) {
            dp[x] = (dp[x] + dp[x - c]) % MOD;
        }
    }
    return dp[X];
}
```

6. Teoría de números

6.1. Cribas

Criba de primos con optimizaciones básicas

TIME	SPACE
$O(n \log \log n)$	$O(n)$

Find all the prime numbers in $[1, n]$. This code uses the next optimizations:

- Sieving till root
- Sieving by the odd numbers only

teoria_de_numeros/sieve_with_basic_optimizations.cpp

```cpp
#include <bits/stdc++.h>
using namespace std;

vector<bool> sieve(int n) {
    vector<bool> is_prime(n + 1, true);
    is_prime[0] = is_prime[1] = false;
    for (int i = 3; i * i <= n; i += 2) {
        if (is_prime[i]) {
            for (int j = i * i; j <= n; j += 2 * i)
                is_prime[j] = false;
        }
    }
    return is_prime;
}
```

6.2. Euclid's Algorithm

The algorithm is based on the formula

$$\gcd(a,b) = \begin{cases} a & b = 0 \\ \gcd(b, a \bmod b) & b \neq 0 \end{cases}$$

6.2.1. Extended Euclid's Algorithm:

Euclid's algorithm can also be extended so that it gives integers x and y for which

$$ax + by = \gcd(a,b)$$

Iterative version

TIME

$O(\log \min(a,b))$

teoria_de_numeros/extended_euclid_algorithm.cpp

```cpp
int gcd(int a, int b, int& x, int& y) {
    x = 1, y = 0;
    int x1 = 0, y1 = 1, a1 = a, b1 = b;
    while (b1) {
        int q = a1 / b1;
        tie(x, x1) = make_tuple(x1, x - q * x1);
        tie(y, y1) = make_tuple(y1, y - q * y1);
```

```cpp
        tie(a1, b1) = make_tuple(b1, a1 - q * b1);
    }
    return a1;
}
```

6.3. Euler's Theorem

Euler's totient function $\varphi(n)$ gives the number of integers between $1, \ldots, n$ that are coprime to n.

Any value of $\varphi(n)$ can be calculated from the prime factorization of n using the formula

$$\varphi(n) = \prod_{i=1}^{k} p_i^{\alpha_i - 1}(p_i - 1)$$

> **Teorema 6.3.1 (Euler's theorem)**
>
> For all positive coprime integers x and m
>
> $$x^{\varphi(m)} \bmod m = 1$$

6.4. Solving Equations

We can efficiently solve a Diophantine equation by using the extended Euclid's algorithm which gives integers x and y that satisfy the equation

$$ax + by = \gcd(a, b)$$

A Diophantine equation can be solved exactly when c is divisible by $\gcd(a, b)$.

If a pair (x, y) is a solution, then also all pairs

$$\left(x + \frac{kb}{\gcd(a, b)}, y - \frac{ka}{\gcd(a, b)}\right)$$

are solutions, where k is any integer.

7. Combinatoria

7.1. Binomial Coefficients

> **Nota**
>
> For $n < k$ the value of $\binom{n}{k}$ is assumed to be zero.

Basic Improved Implementation

```cpp
int C(int n, int k) {
    double res = 1;
    for (int i = 1; i <= k; ++i)
        res = res * (n - k + i) / i;
    return (int)(res + 0.01);
}
```

Table of binomial coefficients using Pascal's Triangle identity

```cpp
const int maxn = ...;
int C[maxn + 1][maxn + 1];
C[0][0] = 1;
for (int n = 1; n <= maxn; ++n) {
    C[n][0] = C[n][n] = 1;
    for (int k = 1; k < n; ++k)
        C[n][k] = C[n - 1][k - 1] + C[n - 1][k];
}
```

8. Strings

8.1. Trie

Estructura

strings/trie.cpp

```cpp
const int K = 26;

struct Vertex {
    int next[K];
    bool output = false;

```

```cpp
    Vertex() {
        fill(begin(next), end(next), -1);
    }
};

vector<Vertex> trie(1);
```

Agregar una cadena

strings/trie.cpp

```cpp
void add_string(string const& s) {
    int v = 0;
    for (char ch : s) {
        int c = ch - 'a';
        if (trie[v].next[c] == -1) {
            trie[v].next[c] = trie.size();
            trie.emplace_back();
        }
        v = trie[v].next[c];
    }
    trie[v].output = true;
}
```

8.2. String hashing

hash-doble

strings/Hash_Doble.cpp

```cpp
struct Hash {//doble hashing
    vector<ll> h1, h2;
    vector<ll> p1, p2;
    ll p=31;//tomamos esta p para evitar colisiones
    Hash(string &s) {
        int n = s.size();
        h1.resize(n+1, 0);
        h2.resize(n+1, 0);
        p1.resize(n+1, 1);
        p2.resize(n+1, 1);
//hacemos doble hasheo para evitar colisiones
        for (int i = 0; i < n; i++) {
            p1[i+1] = (p1[i] * p) % MOD;//a uno le aplicamos modulo
            p2[i+1] = (p2[i] * p);//el otro ocupamos el modulo implicito de ll
//uno con mod 10e9+7 y otro con 2e64 que es LLONG_MAX
            int val = s[i] - 'a' + 1;//hasheamos con el asci

            h1[i+1] = (h1[i] * p + val) % MOD;
            h2[i+1] = (h2[i] * p + val);
        }
    }

    pair<ll,ll> get(ll l, ll r) {//comparamos los 2 hashing
        ll x1 = (h1[r+1] - h1[l] * p1[r-l+1] % MOD + MOD) % MOD;
        ll x2 = (h2[r+1] - h2[l] * p2[r-l+1]);
        return {x1, x2};
    }
};
Hash hs(s);//inicializar
hs.get(i,j);//lugar de sonde se compara
if(hs.get(l,m)==hs.get(i,j))//si 2 cadenas son iguales
```

8.3. KMP — Knuth-Morris-Pratt

KMP — Knuth-Morris-Pratt

TIME	SPACE
$O(n+m)$	$O(m)$

Cuenta ocurrencias de pat en txt. buildPhi construye el arreglo de fallos: phi[j] = prefijo propio más largo de pat[0..j] que es sufijo. Al fallar retrocede a phi[i-1] en lugar de reiniciar.

strings/kmp.cpp

```cpp
vll buildPhi(const vll& pat) {
    ll m = pat.size();
    vll phi(m, 0);
    for (ll j = 1, i = 0; j < m; j++) {
        while (i > 0 && pat[i] != pat[j]) i = phi[i - 1];
        if (pat[i] == pat[j]) i++;
        phi[j] = i;
    }
    return phi;
```

```cpp
}

ll KMP(const vll& pat, const vll& txt) {
    ll m = pat.size(), n = txt.size();
    if (m == 0) return 0;
    vll phi = buildPhi(pat);
    ll matches = 0;
    for (ll i = 0, j = 0; j < n; j++) {
        while (i > 0 && pat[i] != txt[j]) i = phi[i - 1];
        if (pat[i] == txt[j]) i++;
        if (i == m) { matches++; i = phi[i - 1]; }
    }
    return matches;
}
```

9. Manipulación de bits

9.1. Máscara de bits

9.1.1. Manipulación individual de bits

Establece en 1 el *k*-ésimo bit de `x`:

```cpp
#define ON(x, k) ((x) | (1LL << (k)))
```

Establece en 0 el *k*-ésimo bit de `x`:

```cpp
#define OFF(x, k) ((x) & ~(1LL << (k)))
```

Invierte el *k*-ésimo bit de `x`:

```cpp
#define TOGGLE(x, k) ((x) ^ (1LL << (k)))
```

Comprueba si el *k*-ésimo bit de `x` **es 1**:

```cpp
#define CHECK(x, k) ((x) & (1LL << (k)))
```

9.1.2. Working with the rightmost 1-bit

`x & -x`: Isolates the rightmost set bit (keeps only the least significant bit that is 1, sets all others to 0).

`x & (x - 1)`: Removes the rightmost set bit from `x` (turns the rightmost 1 to 0).

`x | (x - 1)`: Sets all trailing zeros of `x` to 1.

9.1.3. Creating useful bit masks

`(1 << n) - 1`: Creates a bit mask with the first n bits set to 1.

9.1.4. Common bit manipulation tricks

- A positive number `x` is a power of two exactly when `x & (x - 1) = 0`.

`x ^ A ^ B`: Determines which is not the current value of `x` between `A` and `B`.

9.1.5. Bit operations as math alternatives

`x << k`: equivalent to `x * pow(2, k)`

`x >> k`: equivalent to `x / pow(2, k)` (integer division)

`x & ((1 << k) - 1)`: equivalent to `x % pow(2, k)`

`A + B`: equivalent to `(A ^ B) + 2 * (A & B)` or `(A | B) + (A & B)`

9.1.6. GNU C++ compiler built-in functions

`__builtin_clz(x)`: the number of leading zeros (zeros on the left side of the bit representation).

`__builtin_ctz(x)`: the number of trailing zeros (zeros on the right side of the bit representation).

`__builtin_popcount(x)`: the number of ones in the bit representation.

`__builtin_parity(x)`: the parity (even or odd) of the number of ones in the bit representation.

Example of use

```cpp
int x = 5328; // 00000000000000000001010011010000
cout << __builtin_clz(x) << "\n";      // 19
cout << __builtin_ctz(x) << "\n";      // 4
cout << __builtin_popcount(x) << "\n"; // 5
cout << __builtin_parity(x) << "\n";   // 1
```

> **Nota**
>
> The above functions only support `int` numbers, but there are also `long long` versions of the functions available with the suffix `ll` (e.g. `__builtin_popcountll(x)`).

9.2. Representing Sets

Every subset of a set $\{0, 1, 2, \ldots, n - 1\}$ can be represented as an n bit integer whose one bits indicate which elements belong to the subset.

Goes through the subsets with exactly k elements

```
for (int b = 0; b < (1<<n); b++) {
    if (__builtin_popcount(b) == k) {
        // process subset b
    }
}
```

Goes through the subsets of a set x

```
int b = 0;
do {
    // process subset b
} while (b=(b-x)&x);
```

The idea is that the formula $b - x$ detects the rightmost one bit in x that is zero in b. This bit becomes one and all bits after it become zero. Then the and operation ensures that the resulting value is a subset of x. Note that $b - x$ equals $-(x - b)$ so we can think that we first remove all one bits that appear in b and then invert the value and add one.

9.3. Bitsets

The **bitset** structure corresponds to an array whose each value is either 0 or 1.

Creates a bitset of 10 elements

```
bitset<10> s;
s[1] = 1;
s[3] = 1;
s[4] = 1;
s[7] = 1;
cout << s[4] << "\n"; // 1
cout << s[5] << "\n"; // 0
```

Returns the number of one bits in the bitset

```
cout << s.count() << "\n"; // 4
```

Bit operations can be directly used to manipulate bitsets

```
bitset<10> a, b;
// ...
bitset<10> c = a&b;
bitset<10> d = a|b;
bitset<10> e = a^b;
```

Recorrer bitset

```
for (int i = 0; i < (int)bs.size(); ++i) {
    cout << "bit " << i << ": " << bs[i] << "\n";
}
```

Mostrar máscara como binario

```
cout << "maskNum = " << bitset<8>(maskNum) << '\n';
```

10. Definiciones y resultados útiles

10.1. Teoría de números

10.1.1. Primos

- Para todo entero $n > 1$, existe una factorización prima única

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$

- La función contadora de primos $\pi(n)$ da el *número de primos hasta n*:

$$\pi(n) \approx \frac{n}{\ln(n)}$$

- Número de divisores de un entero n:

$$\tau(n) = \prod_{i=1}^{k} (\alpha_i + 1)$$

- Suma de divisores de un entero n:

$$\sigma(n) = \prod_{i=1}^{k} \left(1 + p_i + \ldots + p_i^{\alpha_i}\right) = \prod_{i=1}^{k} \frac{p_i^{\alpha_i+1} - 1}{p_i - 1}$$

- Producto de divisores de un entero n:

$$n^{\frac{\tau(n)}{2}}$$

10.1.2. Aritmética modular

Definición 10.1.1 (Modular multiplicative inverse)

The *modular multiplicative inverse* of x with respect to m is a value $\mathrm{inv}_m(x)$ such that

$$x \cdot \mathrm{inv}_m(x) \bmod m = 1$$

If m is prime

$$\mathrm{inv}_m(x) = x^{m-2}$$

- Factorial inverso modular: $\mathrm{inv}_p((x-1)!) \equiv \left(x \cdot \mathrm{inv}_p(x!)\right) \bmod p$

10.2. Combinatoria

10.2.1. Números de Fibonacci

$$F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$$

10.2.2. Números de Catalán

10.2.3. Coeficientes binomiales

- Pascal's Triangle:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

- Symmetry rule:

$$\binom{n}{k} = \binom{n}{n-k}$$

- Factoring in:

$$\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}$$

- Sum over k:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

- Sum over n:

$$\sum_{m=0}^{n} \binom{m}{k} = \binom{n+1}{k+1}$$

- Sum over n and k:

$$\sum_{k=0}^{m} \binom{n+k}{k} = \binom{n+m+1}{m}$$

- Sum of the squares:

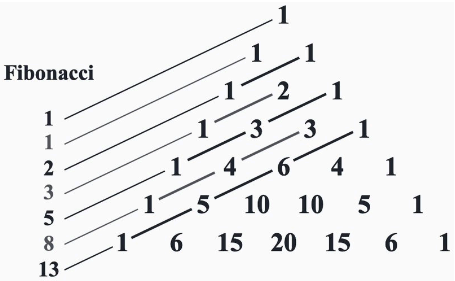
$$\binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2 = \binom{2n}{n}$$

- Weighted sum:

$$1\binom{n}{1} + 2\binom{n}{2} + \cdots + n\binom{n}{n} = n2^{n-1}$$

- Connection with the Fibonacci numbers:

$$\binom{n}{0} + \binom{n-1}{1} + \cdots + \binom{n-k}{k} + \cdots + \binom{0}{n} = F_{n+1}$$



10.2.4. Estrellas y barras

Teorema 10.2.1

El número de formas de colocar n objetos idénticos en k cajas etiquetadas es

$$\binom{n+k-1}{n}$$

Utilizando este resultado podemos contar el número de sumas de k enteros acotados inferiormente, i.e, el número de soluciones para la ecuación

$$x_1 + x_2 + \cdots + x_k = n$$

con $x_i \geq a_i$.

La solución es:

$$\binom{n - (a_1 + a_2 + \cdots + a_k) + k - 1}{n}$$

Cuando los x_i están acotados superiormente, podemos usar el principio de inclusión-exclusión para contar el número de soluciones.

10.2.5. Principio de inclusión-exclusión

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n-1} |A_1 \cap A_2 \cap \cdots \cap A_n|$$

10.2.6. Desarreglos (subfactorial)

$$D_n = \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)! \approx \frac{n!}{e}$$

$$D_n = \begin{cases} 0 & \text{si } n = 1 \\ 1 & \text{si } n = 2 \\ (n-1)\left(D_{n-2} + D_{n-1}\right) & \text{si } n \geq 3 \end{cases}$$

$$D_n = nD_{n-1} + (-1)^n \quad \text{para} \quad n \geq 1$$

10.3. Primeros términos de algunas sucesiones básicas

n	p_n	F_n	C_n	2^n	$n!$	$!n$
0	—	0	1	1	1	1
1	2	1	1	2	1	0
2	3	1	2	4	2	1
3	5	2	5	8	6	2
4	7	3	14	16	24	9
5	11	5	42	32	120	44
6	13	8	132	64	720	265
7	17	13	429	128	5040	1854
8	19	21	1430	256	40320	14833
9	23	34	4862	512	362880	133496
10	29	55	16796	1024	3628800	1334961
11	31	89	58786	2048	39916800	14684570
12	37	144	208012	4096	479001600	176432560
13	41	233	742900	8192	6227020800	2290792932
14	43	377	2674440	16384	87178291200	32071101049
15	47	610	9694845	32768	1307674368000	481066515734

11. Utilidades

11.1. STL containers

11.1.1. Deques

A ***deque*** is a dynamic array that can be efficiently manipulated at both ends of the structure.

```
deque<int> d;
d.push_back(5); // [5]
d.push_back(2); // [5,2]
d.push_front(3); // [3,5,2]
d.pop_back(); // [3,5]
d.pop_front(); // [5]
```

The operations of a deque also work in $O(1)$ average time. Deques should be used only if there is a need to manipulate both ends of the array.

11.1.2. Set Structures

The basic operations of sets are element insertion, search and removal. Sets are implemented so that all the above operations are efficient.

11.1.2.1. Sets and Multisets

Basic functions

- `insert` adds an element to the set
- `count` returns the number of occurrences of an element in the set
- `erase` removes an element from the set
- `find(x)` returns an iterator that points to an element whose value is x. However, if the set does not contain x, the iterator will be `end()`.

Use of find

```
auto it = s.find(x);
if (it == s.end()) {
    // x is not found
}
```

Multisets: A *multiset* is a set that can have several copies of the same value. C++ has the structures `multiset` and `unordered_multiset`.

```
multiset<int> s;
s.insert(5);
s.insert(5);
```

```
s.insert(5);
cout << s.count(5) << "\n"; // 3
```

The function erase removes all copies of a value from a multiset:

```
s.erase(5);
cout << s.count(5) << "\n"; // 0
```

Often, only one value should be removed, which can be done as follows:

```
s.erase(s.find(5));
cout << s.count(5) << "\n"; // 2
```

Note that the functions **`count`** and **`erase`** have an additional $O(k)$ factor where k is the number of elements counted/removed. In particular, it is *not* efficient to count the number of copies of a value in a multiset using the **`count`** function.

11.1.3. Priority Queues

Insertion and removal take $O(\log n)$ time, and retrieval takes $O(1)$ time.

If we only need to efficiently find minimum or maximum elements, it is a good idea to use a priority queue instead of a set or multiset.

By default, the elements in a C++ priority queue are sorted in decreasing order, and it is possible to find and remove the largest element in the queue.

```
priority_queue<int> q;
q.push(3);
q.push(5);
q.push(7);
q.push(2);
cout << q.top() << "\n"; // 7
q.pop();
cout << q.top() << "\n"; // 5
q.pop();
q.push(6);
cout << q.top() << "\n"; // 6
q.pop();
```

If we want to create a priority queue that supports *finding and removing the smallest element*, we can do it as follows:

```
priority_queue<int, vector<int>, greater<int>> q;
```

11.2. Strings

Method/Syntax	Description	Complexity	Example
`s.insert(pos, str)`	Inserts string at position	O(n+m)	`s.insert(2, "abc");`
`s.erase(pos, len)`	Removes len characters from pos	O(n)	`s.erase(2, 3);`
`s.substr(pos, len)`	Returns substring from pos with len chars	O(len)	`string sub = s.substr(2, 5);`
`s.find(str, pos)`	Finds first occurrence of str from pos	O(n*m)	`size_t pos = s.find("abc");`
`s.rfind(str, pos)`	Finds last occurrence of str before pos	O(n*m)	`size_t pos = s.rfind("abc");`
`s.find_first_of(chars)`	Finds first occurrence of any char in chars	O(n*m)	`s.find_first_of("aeiou");`
`s.find_last_of(chars)`	Finds last occurrence of any char in chars	O(n*m)	`s.find_last_of("aeiou");`
`s.find_first_not_of(chars)`	Finds first char not in chars	O(n*m)	`s.find_first_not_of(" \t");`
`s.find_last_not_of(chars)`	Finds last char not in chars	O(n*m)	`s.find_last_not_of(" \t");`
`s.replace(pos, len, str)`	Replaces len chars at pos with str	O(n+m)	`s.replace(2, 3, "xyz");`
`s.compare(str)`	Lexicographically compares strings	O(min(n,m))	`int cmp = s.compare("abc");`
`s.c_str()`	Returns C-style null-terminated string	O(1)	`const char* cstr = s.c_str();`
`s.data()`	Returns pointer to character array	O(1)	`char* ptr = s.data();`
`s.reserve(n)`	Reserves capacity for n characters	O(1)	`s.reserve(1000);`
`s.resize(n, c)`	Changes size to n, filling with c if needed	O(n)	`s.resize(10, 'x');`
`s.clear()`	Removes all characters	O(n)	`s.clear();`
`s.swap(other)`	Swaps contents with another string	O(1)	`s.swap(t);`
`getline(cin, s)`	Reads entire line including spaces	O(n)	`getline(cin, s);`
`getline(cin, s, delim)`	Reads until delimiter character	O(n)	`getline(cin, s, ',');`
`stoi(s)` / `stol(s)` / `stoll(s)`	Converts string to int/long/long long	O(n)	`int num = stoi("123");`
`stof(s)` / `stod(s)`	Converts string to float/double	O(n)	`double d = stod("3.14");`
`to_string(num)`	Converts number to string	O(log num)	`string s = to_string(42);`
`isalpha(c)` / `isdigit(c)`	Checks if char is letter/digit	O(1)	`if (isalpha(s[i])) { ... }`
`islower(c)` / `isupper(c)`	Checks if char is lowercase/uppercase	O(1)	`if (islower(s[i])) { ... }`
`tolower(c)` / `toupper(c)`	Converts char to lowercase/uppercase	O(1)	`s[i] = tolower(s[i]);`
`isspace(c)`	Checks if char is whitespace	O(1)	`if (isspace(s[i])) { ... }`
`isalnum(c)`	Checks if char is alphanumeric	O(1)	`if (isalnum(s[i])) { ... }`

11.3. STL Algorithm

Function/Syntax	Description	Complexity	Example

`sort(begin, end)`	Sorts elements in ascending order	O(n log n)	`sort(v.begin(), v.end())`
`stable_sort(begin, end)`	Sorts maintaining relative order of equal elements	O(n log n)	`stable_sort(v.begin(), v.end())`
`reverse(begin, end)`	Reverses the order of elements	O(n)	`reverse(v.begin(), v.end())`
`binary_search(begin, end, val)`	Searches for value in sorted range (returns bool)	O(log n)	`binary_search(v.begin(), v.end(), x)`
`lower_bound(begin, end, val)`	Finds first element ≥ val	O(log n)	`lower_bound(v.begin(), v.end(), x)`
`upper_bound(begin, end, val)`	Finds first element > val	O(log n)	`upper_bound(v.begin(), v.end(), x)`
`equal_range(begin, end, val)`	Returns pair of iterators [lower_bound, upper_bound]	O(log n)	`equal_range(v.begin(), v.end(), x)`
`min_element(begin, end)`	Finds iterator to minimum element	O(n)	`min_element(v.begin(), v.end())`
`max_element(begin, end)`	Finds iterator to maximum element	O(n)	`max_element(v.begin(), v.end())`
`minmax_element(begin, end)`	Returns pair {min_iter, max_iter}	O(n)	`minmax_element(v.begin(), v.end())`
`next_permutation(begin, end)`	Generates next lexicographic permutation	O(n)	`next_permutation(v.begin(), v.end())`
`prev_permutation(begin, end)`	Generates previous lexicographic permutation	O(n)	`prev_permutation(v.begin(), v.end())`
`unique(begin, end)`	Removes consecutive duplicate elements	O(n)	`unique(v.begin(), v.end())`
`count(begin, end, val)`	Counts occurrences of a value	O(n)	`count(v.begin(), v.end(), x)`
`count_if(begin, end, pred)`	Counts elements satisfying condition	O(n)	`count_if(v.begin(), v.end(), [](int x){return x%2==0;})`
`find(begin, end, val)`	Finds first occurrence of value	O(n)	`find(v.begin(), v.end(), x)`
`find_if(begin, end, pred)`	Finds first element satisfying condition	O(n)	`find_if(v.begin(), v.end(), [](int x){return x>5;})`
`rotate(begin, middle, end)`	Rotates elements (middle becomes new begin)	O(n)	`rotate(v.begin(), v.begin()+k, v.end())`
`random_shuffle(begin, end)`	Shuffles elements randomly	O(n)	`random_shuffle(v.begin(), v.end())`
`shuffle(begin, end, rng)`	Shuffles with specific random generator	O(n)	`shuffle(v.begin(), v.end(), rng)`
`fill(begin, end, val)`	Fills range with a value	O(n)	`fill(v.begin(), v.end(), 0)`
`iota(begin, end, val)`	Fills with consecutive values starting from val	O(n)	`iota(v.begin(), v.end(), 1)`
`accumulate(begin, end, init)`	Sums elements (requires `<numeric>`)	O(n)	`accumulate(v.begin(), v.end(), 0)`
`partial_sum(begin, end, out)`	Calculates partial sums	O(n)	`partial_sum(v.begin(), v.end(), prefix.begin())`
`merge(begin1, end1, begin2, end2, out)`	Merges two sorted ranges	O(n+m)	`merge(a.begin(), a.end(), b.begin(), b.end(), c.begin())`
`inplace_merge(begin, middle, end)`	Merges two sorted parts of same range	O(n log n)	`inplace_merge(v.begin(), v.begin()+k, v.end())`
`set_union(begin1, end1, begin2, end2, out)`	Union of two sorted sets	O(n+m)	`set_union(a.begin(), a.end(), b.begin(), b.end(), c.begin())`

`set_intersection(begin1, end1, begin2, end2, out)`	Intersection of two sorted sets	O(n+m)	`set_intersection(a.begin(), a.end(), b.begin(), b.end(), c.begin())`
`set_difference(begin1, end1, begin2, end2, out)`	Difference of two sorted sets	O(n+m)	`set_difference(a.begin(), a.end(), b.begin(), b.end(), c.begin())`
`nth_element(begin, nth, end)`	Partitions so nth element is in correct position	O(n) avg	`nth_element(v.begin(), v.begin()+k, v.end())`
`partition(begin, end, pred)`	Partitions elements by predicate	O(n)	`partition(v.begin(), v.end(), [](int x) {return x%2==0;})`
`is_sorted(begin, end)`	Checks if range is sorted	O(n)	`is_sorted(v.begin(), v.end())`
`is_permutation(begin1, end1, begin2)`	Checks if one range is permutation of another	$O(n^2)$	`is_permutation(a.begin(), a.end(), b.begin())`

11.3.1. Useful Combinations:

```
// Sort and remove duplicates
sort(all(v));
v.erase(unique(all(v)), v.end());
```